Response to the supplied external review

Certified global annihilation barriers in an infinite XY lattice

Pre-submission revision, 20 October 2026

The supplied comments identify useful improvements in motivation and geometric accessibility. The revision addresses all six substantive and editorial points below. It preserves the theorem statements, the admissible strong-energy-space path class, the certified barrier enclosure, and the coupling budget. The favorable assessment in the report is not used as evidence of mathematical correctness or bibliographic priority. This response concerns the review supplied by the author before submission; it is not a response to an identified SIADS editorial decision.

1 Motivation for the energy-space gradient

Comment. Explain why the chosen Hilbert metric and flow have a physical or mathematical role beyond functional-analytic convenience.

Change. Section 2 now contains a subsection entitled "Dissipation metric and dynamical interpretation." It starts from the Rayleigh dissipation potential

$$\mathcal{R}(v) = \tfrac{1}{2} \sum_e |D_e v|^2 = \tfrac{1}{2} \|v\|_{\mathcal{X}}^2.$$

Minimizing $\mathcal{R}(v) + DE_b(u)[v]$ gives the weak gradient equation and the identity $\dot{E}_b = -\sum_e |D_e \dot{u}|^2$. The metric therefore corresponds to bond dissipation of phase-difference velocities. The subsection derives the finite-array current-conservation interpretation for overdamped resistively shunted Josephson junctions, with equal shunt conductances, fixed boundary phases, no injected current and no noise. Equation (13) of Yoon–Choi–Kim (2000), already cited in the article, supports this interpretation. Its additional fluctuating-twist equations are explicitly excluded.

The infinite lattice is formulated directly in the energy space; the revision does not assume a thermodynamic limit of a driven stochastic circuit model. It also distinguishes the static variational barrier from its choice of dissipative relaxation law. The same energy and path class define the same barrier if mobility is changed, while orbit and rate questions require additional analysis. The phenomenon certified here is pinned-dipole annihilation and its positive escape barrier, not a proof of the thermodynamic BKT phase transition.

2 A geometric schematic

Comment. Illustrate exact saddle caps, cutting planes and coordinate-face coverage.

Change. Figure 1, placed in Section 3, supplies two complementary views. The reaction-coordinate panel displays the minimum, both saddles, exact cutting lines and the selected upper-path mechanism. The second panel displays a slice of transverse fibers at a coordinate face and the strict inclusion that makes a low-energy part of that face internal to another chart. The text connects the diagram to exact-plane lifting and to the complete boundary ledger.

The figure is labeled as conceptual: its positions, chart count and proportions are not certified numerical geometry. In particular, the infinite-dimensional fiber is not replaced by the drawn slice. The adjacent text explains that the fiber minimizer must fit in the local cap; strong convexity then controls the entire fiber. The image does not substitute overlap of pictures for the complete-face inclusion test. The Python source, vector PDF/SVG, preview PNG, caption and alt text are supplied with the editable sources.

3 Effective localization and practical computation

Comment. Identify specific cancellations or averaging methods that could reduce the very large truncation budgets.

Change. Section 7 now separates the two cost mechanisms. The proved error is C_E/R, so a target $\varepsilon/4$ requires $R \geq 4C_E/\varepsilon$. The present $C_E \simeq 1.60 \times 10^{17}$ is large, but the dependence on inverse accuracy is linear. The exponential obstruction is a complete cubical search in the number of variables, itself of order R^2.

The concrete next steps are data-dependent core sums of the nonlinear source and separate exterior bounds, followed by leading-multipole subtraction before absolute values. An exact kernel-difference identity shows where this cancellation would enter. An improved rate or useful constant still requires quantitative kernel differences, source moments, and uniform control along the path and admitted coupling class. Neutral vortex charge alone does not establish the required moments or their cancellations. No unproved multiphase averaging is presented as a solution: it need not preserve real lifts, charges, or the energy ceiling. The revision retains the low-rank atlas plus analytic exterior as the practical architecture, without claiming a new feasible global solver or a new barrier computation.

4 Abstract readability

Comment. Shorten long sentences and separate the saddle-height identity from its numerical value.

Change. The abstract has been rewritten with shorter sentences. The barrier identity and the strict numerical enclosure are now separate sentences. The path class, reusable conditions, effective tail and robustness are still stated, including the unresolved perturbed minimax-realization question. The revised abstract contains 219 words, below the 250-word limit.

5 Code availability and archival deposit

Comment. Deposit the code and verification archives permanently and replace the draft availability statement by the verified identifier.

Change and policy clarification. We agree with the goal of a durable, versioned deposit. The revised availability statement identifies the actual frozen supplementary archives, index and hashes, and states precisely that the separate public deposit has not yet occurred. No URL or DOI has been invented.

The current SIADS instructions call for indexed supplements and a frozen version of submitted code; we found no general separate-repository DOI deadline before acceptance. The proposed archival deposit is therefore retained as an author action, not represented as a verified universal prerequisite. A specific editor or funder may impose additional conditions. Once an actual public deposit exists, its resolving version-specific identifier should be inserted and checked.

Official sources checked on 20 October 2026:

- `https://epubs.siam.org/journal/siads/instructions-for-authors`
- `https://epubs.siam.org/journal-authors`
- `https://epubs.siam.org/artificial-intelligence`

No public posting or journal submission was performed in this revision.

6 Dependent bond angles in the perturbation bound

Comment. Explain why compatibility constraints among bond angles do not invalidate the lower and upper enclosures for W_b.

Change. The proof of Lemma 6.1 now writes the bondwise inequality

$$2\min(b_e, 0) \le b_e(1 - \cos D_e u) \le 2\max(b_e, 0).$$

It holds for every phase field before summing. Absolute summability of b permits summing it on the infinite lattice. Hence the constraints between increments cannot invalidate the enclosures. Their endpoints need not be attainable minima or maxima; the exact width $2\|b\|_1$ of this enclosing interval is an upper bound on the actual oscillation. No strict inequality or independent choice of bond angles is assumed.

Verification scope of this revision

The changes clarify interpretation and exposition and add a schematic; they do not alter the archived global engine or its frozen input arrays. The compiled files, cross-references, figure generation, supplementary index and exact-rational consequence checkers were rechecked for this revision. The earlier fresh global execution record remains identified by its original hashes. The large global chain was not rerun merely for these editorial changes, and the new schematic is not presented as its output. Author metadata, personal review and any public archival deposit remain separate pre-submission actions.